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### **BRACKET, SYSTEM AND METHOD FOR HANGING ARTICLES UNDER A STRUCTURE**

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#### **Abstract**

The present invention is directed towards a system for hanging an article under a structure, the system comprising a structure comprising at least one plank; an article; and at least one bracket comprising a structure engaging portion structured and arranged to contact a top surface of at least one plank; a bottom extension portion structured and arranged to be coupled to the structure engaging portion and having a length that extends below a bottom surface of the plank; and an article engaging portion structured and arranged to be coupled to the article. Also disclosed are methods of hanging an article underneath a structure, and a bracket.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/049,671, filed on Oct. 26, 2022, now allowed, which claims the benefit of U.S. Provisional Patent Application Ser. No. 63/271,786, filed on Oct. 26, 2021, which is incorporated herein by reference.

### FIELD OF THE INVENTION

[0002] The present invention is directed towards brackets, systems and methods for hanging articles under a structure.

### BACKGROUND INFORMATION

[0003] Elevated structures are used for many purposes. For example, architectural structures provide support for roofs and overhangs as well as flooring. However, hanging articles from such structures is often done using methods that can damage the structure. For example, hanging materials underneath a structure typically involves the use of fasteners that are driven into the structure and leave damage to the structure when they are removed. A system for hanging an article under a structure that does not damage the structure is desired.

### SUMMARY OF THE INVENTION

[0004] The present invention is directed to a system for hanging an article under a structure, the system comprising a structure comprising at least one plank; an article; and at least one bracket comprising a structure engaging portion structured and arranged to contact a top surface of at least one plank; a bottom extension portion structured and arranged to be coupled to the structure engaging portion and having a length that extends below a bottom surface of the plank; and an article engaging portion structured and arranged to be coupled to the article.

[0005] The present invention is also directed to a method of hanging an article underneath a structure using the system of the present invention, wherein the method comprises inserting the bottom extension portion of the bracket between two planks until the structure engaging portion contacts a top surface of the structure; and coupling the article engaging portion with the article.

[0006] The present invention is also directed to a bracket comprising a structure engaging portion structured and arranged to contact a top surface of at least one plank; a bottom extension portion structured and arranged to be coupled to the structure engaging portion and having a length that extends below a bottom surface of the plank; an article engaging portion structured and arranged to be coupled to the article; and a releasable bottom holder structured and arranged to be positioned onto the bottom extension portion to make contact with the bottom surface of the plank.

[0007] The present invention is further directed to a bracket comprising a structure engaging portion structured and arranged to contact a top surface of at least one plank; a bottom extension portion structured and arranged to be coupled to the structure engaging portion and having a length that extends below a bottom surface of the plank; an article engaging portion structured and arranged to be coupled to the article; and an extension arm comprising a movable jaw coupled with the bottom extension portion of the bracket, wherein the extension arm and movable jaw are structured and arranged to position the movable jaw underneath the bottom surface of the plank contacted by the structure engaging portion of the bracket and extend the movable jaw to contact the bottom surface of the plank.

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## Description

## BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. **1** is an isometric side view of an exemplary system of the present invention that includes two brackets to hang a swing.

[0009] FIG. **2** is an isometric side view of an exemplary system of the present invention that includes one bracket to hang a potted plant.

[0010] FIG. **3** is an isometric side view of an exemplary system of the present invention that includes one bracket to hang a birdhouse.

[0011] FIG. **4** is an isometric side view of two exemplary brackets installed in a deck floor having rubber stoppers on either side of the bracket.

[0012] FIG. **5** is a side view of an exemplary system of the present invention having a bracket having an extension arm and movable jaw installed onto a plank of a structure.

[0013] FIG. **6** is a side view of an exemplary system of the present invention having a bracket having a releasable bottom holder.

[0014] FIG. **7** is a top isometric view of two exemplary brackets installed in a deck floor having rubber stoppers on either side of the bracket.

[0015] FIG. **8A**, **8B**, and **8C** are profile (**8A**), arial (**8B**), and spin (**8C**) isometric views of an exemplary bracket of the present invention.

[0016] FIG. **9A**, **9B**, and **9C** are profile (**9A**), arial (**9B**), and spin (**9C**) isometric views of an exemplary bracket of the present invention.

[0017] FIG. **10A**, **10B**, and **10C** are profile (**10A**), arial (**10B**), and spin (**10C**) isometric views of an exemplary bracket of the present invention.

[0018] FIG. **11** is a side view of an exemplary system of the present invention having a bracket installed in a structure and supporting an article.

[0019] FIG. **12A**, **12B**, and **12C** are profile (**12A**), spin (**12B**), and alternate profile (**12C**) isometric views of an exemplary bracket of the present invention.

[0020] FIG. **13A**, **13B**, and **13C** are profile (**13A**), spin (**13B**), and alternate profile (**13C**) isometric views of an exemplary bracket of the present invention.

[0021] FIG. **14A** is an isometric view of an exemplary structure.

[0022] FIG. **14B** is a top view of the planks of an exemplary structure.

[0023] FIG. **14C** is an isometric view showing the planks connected to the legs of an exemplary structure.

[0024] FIG. **15** is an isometric view of an exemplary system of the present invention that includes a swing set structure and two brackets that hang a swing.

[0025] FIG. **16** is an isometric view of two planks with crossbars.

[0026] FIG. **17** is an isometric view of two planks with crossbars and an exemplary bracket to be installed between the crossbars.

## DETAILED DESCRIPTION OF THE INVENTION

[0027] As shown in FIGS. **1-7**, disclosed herein is a system **10** for hanging an article **300** under a structure **100**, the system **10** comprising a structure **100** comprising at least one plank **200**, an article **300**, and at least one bracket **400** comprising a structure engaging portion **500** structured and arranged to contact a top surface **210** of at least one plank **200**, a bottom extension portion **600** structured and arranged to be coupled to the structure engaging portion **500** and having a length that extends below a bottom surface **220** of the plank **200**, and an article engaging portion **700** structured and arranged to be coupled to the article **300**.

[0028] The structure **100** is not limited so long as it includes a plank **200**. As used herein, the term “plank” is intended to refer to a support upon which the structure engaging portion **500** may contact. For example, the structure **100** may comprise voids between the planks **200** or within a plank **200** such that the bottom extension portion **600** may extend from the top of the plank through the structure **100** to reach the underside of the structure **100**. The structure **100** may optionally

comprise a plurality of planks **200**, such as at least 2 planks **200**, upon which the structure engaging portion(s) may contact.

[0029] As shown in FIGS. **1-4**, **7**, and **11**, the structure **100** optionally may comprise a deck **110** or other similar structure having a plurality of planks **200** that form the structure **100**. The deck **110** may optionally further include joists, beams, and posts in addition to the planks **200**. Although the structure engaging portion **500** of the bracket **400** is expressed as being structured and arranged to contact the top surface **210** of a plank **200**, the term plank **200** includes any portion of the deck **110** that is used to make the decking floor **120** and may include planks, joists, beams, and posts if they are exposed as part of the decking floor **120**. As shown in FIGS. **1-4**, **7**, and **11**, the deck **110** includes voids between the planks **200** such that the bottom extension portion **600** of the bracket **400** extends from the top of the deck flooring **120** through the deck **110** to reach the underside of the deck **110**.

[0030] As shown in FIGS. **16** and **17**, the structure **100** may optionally further comprise crossmembers **230** in the void between the two planks **200**. The crossmembers **230** may define slots through which the bracket **400** is inserted and may present movement of the bracket **400** in the void between the planks **200**.

[0031] As shown in FIGS. **14A**, **14B**, **14C**, and **15**, the structure **100** may optionally comprise a structure **100** comprising at least one plank **200** and a plurality of legs **160** that support the at least one plank **200**. As shown in FIGS. **14A**, **14B**, **14C**, and **15**, the structure **100** may comprise two planks **200** that form a void between the two planks **200** through which the bracket **400** may be inserted wherein the bottom extension portion **600** of the bracket **400** extends from the top surface of plank **210** to extend past the bottom surface of the plank **220**. Alternatively, the structure **100** may comprise one plank **200** with having one or more apertures (i.e., voids) running the length of the plank **200** through which the bracket **400** may be installed.

[0032] As shown in FIGS. **16** and **17**, the structure **100** may be a swing set **150** that may further comprise the crossmembers **230** as described above that restrict movement of the bracket **400** once installed between the planks **200**. One or more bracket(s) **400** may be installed into the swing set **150** and connected to an article **300**, such as, for example, the swings described below. This configuration may allow for the swing set **150** to be easily adjusted and reconfigured by the user to include different articles **300** supported by the bracket(s) **400** in the swing set **150**.

[0033] The article **300** is not limited and may comprise, for example, a swing, a swinging bench, a hammock, a hanging basket, such as a hanging plant, a floating garden, a birdhouse, a hanging light, or any other article **300** that could be positioned beneath a structure **100**. The swing may comprise a tire swing, a rope swing, a sling swing, a disc swing, a flat swing, a half-bucket swing, a full-bucket swing, a porch swing, a glider, a horse glider, or a web swing.

[0034] According to the present invention, the bracket **400** may comprise any suitable material. For example, the bracket **400** may comprise, metals, such as steel, aluminum, copper, cast iron, wrought iron, titanium, and the like, plastics, such as thermoplastics or thermoset plastics, silicone, wood, wood-plastic composite, stone, rubber, and the like.

[0035] The bracket **400** may comprise a t-bracket comprising two structure engaging portions as shown in FIGS. **1-3**, **5**, **6**, **8A**, **8B**, **8C**, **9A**, **9B**, **9C**, **10A**, **10B**, **10C**, **11**, **12A**, **12B**, **12C**, **13A**, **13B**, **13C**, and **17**. Each of the structure engaging portions may be structured and arranged to contact a top surface **210** of two different planks **200**, such as two adjacent planks **200** with the bottom extension portion **600** of the bracket **400** extending through the structure **100** between the two planks **200**. The width of the structure engaging portions is not limited but should be wide enough to span the distance of the void between the two planks. For example, the structure engaging portions of the bracket **400** may be at least two times greater than the distance of the void between the planks in order to prevent the bracket **400** from falling into the void, supporting the object **300** without the use of fasteners or other mechanisms that damage the plank(s) **200**.

[0036] The structure engaging portion **500** may optionally further comprise a material that is

structured and arranged to restrict movement between the structure engaging portion **500** and the top surface **210** of the plank **200**. The material may comprise a coating added to the structure engaging portion, a pad adhered to the structure engaging portion, an adhesive added to the structure engaging portion, or a coarse surface that may be added or integral to the structure engaging portion **500**, and the material may increase friction between the structure engaging portion **500** and the top surface of the plank **210**. For example, the coating may comprise a coating having an increased co-efficient of friction between the structure engaging portion **500** and top surface **210** of the plank **200**. A non-limiting example includes a silicone resin. The coating may be present over the structure engaging portion **500** alone and/or may be coated over other parts of the bracket **400**, including the entire bracket **400**. The pad may comprise any material that increases the co-efficient of friction between the structure engaging portion **500** and top surface **210** of the plank **200**. For example, the pad may comprise rubber or silicone among other possible materials. The adhesive may be used to adhere the structure engaging portion **500** to the top surface **210** of the plank **200**. For example, the adhesive may comprise a pressure sensitive adhesive. The coarse surface may include a roughened surface of the structure engaging portion **500** or a coarse material that is added to the structure engaging portion. For example, a piece of sandpaper may be adhered to the structure engaging portion **500** in order to increase friction between the structure engaging portion **500** and top surface **210** of the plank **200**.

[0037] As shown in FIGS. **4** and **7**, the system **10** may optionally further comprise one or more rubber stoppers **1000** that may be placed in the void between the planks **200** next to the structure engaging portion **500** of the bracket **400** to prevent the bracket **400** from moving in the void. Although referred to as rubber, the rubber stoppers **1000** may comprise any suitable material, such as, for example, rubber, silicone, cork, wood, plastic, wood-plastic composites, and the like.

[0038] As shown in FIGS. **1-3**, **5**, **6**, **8A**, **8B**, **8C**, **9A**, **9B**, **9C**, **10A**, **10B**, **10C**, **11**, **12A**, **12B**, **12C**, **13A**, **13B**, **13C**, and **17**, the bracket **400** further comprises a bottom extension portion **600** structured and arranged to be coupled to the structure engaging portion **500** and having a length that extends below a bottom surface **220** of the plank **200**. The bottom extension portion **600** is not limited and may comprise various structures so long as the thickness of the bottom extension portion **600** does not exceed the void between the planks **200** that would restrict the ability to extend between the planks **200** to the bottom surface **220** of the planks **200**, and the length is sufficient to extend past the bottom surface **220** of the planks **200**. For example, as shown in FIGS. **1**, **2**, **3**, **8A**, **8B**, **8C**, **11**, the bottom extension portion **600** may comprise a rectangular shape, but the bottom extension portion **600** may also comprise any geometric shape, for example, as shown in FIGS. **12A**, **12B**, **12C**, **13A**, **13B**, and **13C**, and the bottom extension portion **600** may be a unitary piece or a hollow piece.

[0039] As shown in FIG. **6**, the bracket **400** may optionally further comprise a releasable bottom holder **800** that is structured and arranged to be positioned onto the bottom extension portion **600** to make contact with the bottom surface **220** of the plank **200** and secure the bracket **400** in place. The releasable bottom holder **800** is not limited and may comprise any suitable mechanism to be secured to the bottom extension portion **600**. For example, the releasable bottom holder **800** may comprise an aperture through which the bottom extension portion **600** is positioned and a wingnut that can be tightened to secure the releasable bottom holder **800** to the bottom extension portion **600**. In another example, the bottom extension portion **600** may comprise a series of serrations and the releasable bottom holder **800** may comprise a ratchet mechanism to secure the releasable bottom holder **800** to the bottom extension portion **600**. However, in all cases the securing mechanism should avoid causing damage to the structure **100** and only be used to further secure the bracket **400**.

[0040] As shown in FIG. **5**, the bracket **400** may optionally further comprise an extension arm **900** comprising a movable jaw **910** coupled with the bottom extension portion **600** of the bracket **400**, wherein the extension arm **900** and movable jaw **910** are structured and arranged to position the

movable jaw **910** underneath the bottom surface **220** of the plank **200** contacted by the structure engaging portion **500** of the bracket **400** and extend the movable jaw **910** to contact a bottom surface **220** of the plank **200**. The movable jaw **910** may be tightened to secure the bracket **400** to the plank **200** and restrict movement of the bracket **400**. The extension arm **900** comprising the movable jaw **910** may be coupled with the engaging portion or bottom extension portion **600** after the bottom extension portion **600** has been positioned between the planks **200**. Accordingly, the bottom extension portion **600** may be structured and arranged to couple with the extension arm **900** comprising the movable jaw **910**.

[0041] The bracket **400** may optionally further comprise a bottom structure engaging portion **500** releasably coupled to the bottom extension portion **600** of the bracket **400** and structured and arranged to contact a bottom surface **220** of at least one plank **200**. For example, the bottom structure engaging portion **500** may comprise a plate having a center aperture wherein the bottom extension portion **600** can be placed through the aperture and the bottom extension portion **600** may comprise teeth that releasably ratchet with the plate. The plate may be extended over the bottom extension portion **600** until it contacts the bottom surface **220** of the plank **200** and may be ratcheted in place against the bottom surface **220** of the plank **200**.

[0042] The bracket **400** may optionally be free of fasteners that couple the bracket **400** and any plank **200** or of any nail-like points that extend from the structure engaging portion **500** that would contact the top surface **210** of the plank **200**. Accordingly, the bracket **400** of the present invention can be installed without damaging the structure or any planks **200** with fasteners, nails, or any other component that would damage or pierce the structure **100** material(s).

[0043] As shown in FIGS. **1-3**, **5**, **6**, **8A**, **8B**, **8C**, **9A**, **9B**, **9C**, **10A**, **10B**, **10C**, **11**, **12A**, **12B**, **12C**, **13A**, **13B**, **13C**, and **17**, the bracket **400** further comprises an article engaging portion **700** structured and arranged to be coupled to the article **300**. The article engaging portion **700** is not limited and may comprise any suitable structure that could be used to couple to an article **300**. The article engaging portion **700** may comprise an aperture extending through the bottom extension portion **600** of the bracket **400**, and a structure may be connected to the article engaging portion **700** aperture and the article **300**, such as, for example, a ring, an S-hook, a carabiner, or any other suitable structure. Alternatively, the article engaging portion **700** may comprise a hook structure at the bottom of the bottom extension portion **600**, or may comprise a ring, S-hook, carabiner, or any other suitable structure that is built into the bottom extension portion **600**.

[0044] The article engaging portion **700** may be coupled to the article **300** through use of a line between the article engaging portion **700** and article **300**. For example, the line may comprise a rope, a chain, a rubber, a plastic, a cord, or a cable, any combination thereof. The line allows for more flexibility and movement of the system, and may be useful for a swing, a hanging basket, or any other article **300**.

[0045] The system **10** of the present invention may optionally comprise at least two brackets **400**. The two brackets **400** may be positioned on the structure **100** with each of the article engaging members engaged to a portion of the article **300** or a line extending from the article **300**. For example, as shown in FIGS. **1** and **15**, two brackets **400** may be installed on the structure **100**, and each engaging member of the brackets **400** may be coupled to a line extending from a swing to install a swing underneath the structure **100**.

[0046] The present invention is also directed to a method of hanging an article **300** underneath a structure **100** using the system **10** of any of the present invention, wherein the method comprises inserting the bottom extension portion **600** of the bracket **400** between two planks **200** or through an aperture in a plank **200** until the structure engaging portion **500** contacts the top surface **210** of the plank **200**, and coupling the article engaging portion **700** with the article **300**. Multiple brackets **400** may be installed according to this method depending upon the article **300** to be hung underneath the structure **100**.

[0047] For purposes of the detailed description, it is to be understood that the invention may

assume various alternative variations and step sequences, except where expressly specified to the contrary. Moreover, other than in any operating examples, or where otherwise indicated, all numbers such as those expressing values, amounts, percentages, ranges, subranges and fractions may be read as if prefaced by the word “about,” even if the term does not expressly appear. Accordingly, unless indicated to the contrary, the numerical parameters set forth in the following specification and attached claims are approximations that may vary depending upon the desired properties to be obtained by the present invention. At the very least, and not as an attempt to limit the application of the doctrine of equivalents to the scope of the claims, each numerical parameter should at least be construed in light of the number of reported significant digits and by applying ordinary rounding techniques. Where a closed or open-ended numerical range is described herein, all numbers, values, amounts, percentages, subranges and fractions within or encompassed by the numerical range are to be considered as being specifically included in and belonging to the original disclosure of this application as if these numbers, values, amounts, percentages, subranges and fractions had been explicitly written out in their entirety.

[0048] Notwithstanding that the numerical ranges and parameters setting forth the broad scope of the invention are approximations, the numerical values set forth in the specific examples are reported as precisely as possible. Any numerical value, however, inherently contains certain errors necessarily resulting from the standard variation found in their respective testing measurements.

[0049] As used herein, unless indicated otherwise, a plural term can encompass its singular counterpart and vice versa, unless indicated otherwise. For example, although reference is made herein to “a” bracket, “a” plank, and “an” article, a combination (i.e., a plurality) of these components can be used. In addition, in this application, the use of “or” means “and/or” unless specifically stated otherwise, even though “and/or” may be explicitly used in certain instances.

[0050] As used herein, “including,” “containing” and like terms are understood in the context of this application to be synonymous with “comprising” and are therefore open-ended and do not exclude the presence of additional undescribed or unrecited elements, materials, ingredients or method steps. As used herein, “consisting of” is understood in the context of this application to exclude the presence of any unspecified element, ingredient or method step. As used herein, “consisting essentially of” is understood in the context of this application to include the specified elements, materials, ingredients or method steps “and those that do not materially affect the basic and novel characteristic(s)” of what is being described.

[0051] Whereas specific embodiments of the invention have been described in detail, it will be appreciated by those skilled in the art that various modifications and alternatives to those details could be developed in light of the overall teachings of the disclosure. Accordingly, it is therefore to be understood that the foregoing disclosure is merely illustrative of various exemplary aspects of this application and that numerous modifications and variations can be readily made by skilled artisans which are within the spirit and scope of this application and the accompanying claims.

## Claims

1. A method of hanging an article underneath a structure comprising at least two planks using at least one bracket comprising a structure engaging portion structured and arranged to contact a top surface of at least one of the planks, a bottom extension portion structured and arranged to be coupled to the structure engaging portion and having a length that extends below a bottom surface of at least one of the two planks, and an article engaging portion structured and arranged to be coupled to the article, the method comprising: inserting the bottom extension portion of the bracket between the two planks until the structure engaging portion contacts a top surface of at least one of the two planks; and coupling the article engaging portion with the article.
2. The method of claim 1, wherein the bottom extension portion of the bracket has a length that extends below the bottom surface of each of the two planks.

3. The method of claim 1, wherein the structure engaging portion contacts the top surface of each of the two planks.
  4. The method of claim 1, wherein the bracket comprises a t-bracket comprising two structure engaging portions each structured and arranged to contact the top surface of the two planks comprising two different planks.
  5. The method of claim 1, wherein the structure comprises a deck.
  6. The method of claim 1, wherein the structure comprises a swing set.
  7. The method of claim 1, wherein the structure engaging portion of the bracket further comprises a coarse surface that contacts the top surface of the plank.
  8. The method of claim 1, wherein the structure engaging portion of the bracket further comprises rubber or a silicone resin that contacts the top surface of the plank.
  9. The method of claim 1, wherein at least one structure engaging portion of the bracket further comprises a pressure sensitive adhesive that contacts the top surface of the plank.
  10. The method of claim 1, wherein the bracket further comprises an extension arm comprising a movable jaw coupled with the bottom extension portion of the bracket, wherein the extension arm and movable jaw are structured and arranged to position the movable jaw underneath the plank contacted by the structure engaging portion of the bracket and extend the movable jaw to contact a bottom surface of the plank.
  11. The method of claim 1, wherein the bracket further comprises a bottom structure engaging portion releasably coupled to the bottom extension portion of the bracket and structured and arranged to contact a bottom surface of at least one plank.
  12. The method of claim 1, wherein the article engaging portion comprises an aperture extending through the bottom extension portion, a hook, or a carabiner.
  13. The method of claim 1, wherein the article engaging portion comprises an aperture extending through the bottom extension portion.
  14. The method of claim 1, wherein the method comprises using at least two brackets wherein a second bracket comprises a structure engaging portion structured and arranged to contact a top surface of at least one of the planks, a bottom extension portion structured and arranged to be coupled to the structure engaging portion and having a length that extends below a bottom surface of at least one of the two planks, and an article engaging portion structured and arranged to be coupled to the article, and the method further comprises inserting the bottom extension portion of the second bracket between two planks until the structure engaging portion contacts a top surface of at least one of the planks; and coupling the article engaging portion with the article.
  15. The method of claim 1, wherein the brackets are free of fasteners that couple the bracket and any plank.
  16. The method of claim 1, wherein the method further comprises inserting rubber stoppers between the two planks next to the structure engaging portion of the bracket.
  17. The method of claim 1, wherein the article further comprises at least one line that may be coupled with the article engaging portion.
  18. The method of claim 1, wherein the article comprises a swing.
  19. The method of claim 1, wherein the article comprises a swinging bench, a hammock, a hanging basket, a floating garden, a birdhouse, or a hanging light.
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